

Abdul Khurram

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EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (*Expected Graduation: May 2027*)

Relevant Coursework: **Data Structures & Algorithms**; **Algorithm Design & Analysis**

Euclid Mathematics Contest, University of Waterloo

2021

Top 5% Internationally, Certificate of Distinction

WORK EXPERIENCE

Shopify

Sept. – Dec. 2026

Incoming Software Engineer Intern

Toronto, ON, Canada

RBC Royal Bank | Python, Apache Spark, Airflow, SQL, Linux

Jan. 2026 – Present

Software Developer Co-op

Vancouver, BC, Canada

- Built Python/Airflow pipelines processing **50M+ daily transactions** from petabyte-scale Delta Lake, delivering analysis-ready data for downstream ML and analytics workloads.
- Resolved Spark resource bottlenecks through Spark-native API migration and AQE/cluster-memory tuning, cutting **latency 40%**; built cost/performance dashboards.
- Developing **multi-region pipelines** with observability, fault tolerance, idempotent re-runs, and automated failover for critical banking workloads.

Contello.ai | TypeScript/Node.js, AWS, OpenAI, Playwright

Jan. – Dec. 2025

AI Systems & Full-Stack Engineering Intern

Toronto, ON, Canada

- Built TypeScript/Node.js services and async AWS pipelines (SQS, S3) processing **thousands of content jobs/day**, owning features from API design through production monitoring.
- Integrated the **OpenAI Responses API** and an NLP/embedding context pipeline for brand-aware generation, with validation, responsible-AI guardrails, and cost/latency tracking.
- Raised scheduled-post reliability to **99.9%** through retry/backoff, CloudWatch alerting, Playwright E2E/integration tests, and CI/CD release gates.

RESEARCH EXPERIENCE

UBC Emerging Media Lab | Unreal Engine, C++, Python

Sept. – Dec. 2024

Software Developer & Team Lead

Vancouver, BC, Canada

- Led an interdisciplinary CS × Nursing team building an **AI-driven VR clinical simulation** in Unreal/C++, owning requirements, system design, and delivery.
- Designed a **multithreaded C++/Python supervisor** coordinating concurrent AI actors with shared state and real-time protocol enforcement.
- Profiled the inference pipeline and restructured its call graph, reducing **latency 40% and inference costs 90%**.

PROJECTS

Backer | TypeScript, Python, AWS DynamoDB, XGBoost

Hack The Coast 2026

- Architected a DynamoDB data layer (**10+ tables**) with typed repositories, batch operations, and transactional writes for high-throughput feed and messaging workflows.
- Engineered six behavioral and similarity features and trained an **XGBoost ranker** to personalize startup discovery from interaction signals.

Workbase | Next.js/TypeScript, PostgreSQL, AWS Bedrock

Personal Project

- Solo-built** a full-stack platform that ingests GitHub repositories via OAuth, clusters evidence into work themes, and generates artifacts exclusively from **human-approved claims**.
- Engineered a **resource-bounded AWS Bedrock agent** capped at 8 iterations, 16 tool calls, and 120K tokens/run, with automated **credential redaction** before model calls.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, TypeScript/JavaScript, SQL, Bash

AI/ML: XGBoost, OpenAI API, AWS Bedrock, NLP/embeddings, model evaluation

Infrastructure & Data: Apache Spark, Airflow, Delta Lake; AWS (DynamoDB, S3, SQS, CloudWatch); PostgreSQL, Docker, Git, Linux, CI/CD